

Viability Study of Grid Connected PV/Wind/Biomass Hybrid Energy System for a Small Village in Egypt

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Abstract: Motivated by the new feed-in tariffs for electrical power projects generated from renewable energy resources, the present study is investigating the potential of on-grid hybrid solar/wind/Biomass power generation in Egypt. The arrangement is modeled and examined through the environmental and economic prospects. The optimization, sizing and economic assessment of the introduced systems were accomplished using computer package software called HOMER.

A case study area, Monshaet Taher village at Beni-Suef Governorate, Egypt with (29° 1' 17.0718"N, 30° 52' 17.04"E) is placed for economic feasibility in this study. HOMER optimization model plan was designed with annual average solar

According to the World Bank report, Egypt has the third largest inhabitation in Africa, and the third highest gross national income (GNI) [1]. Egypt's national economy has suffered during and after the revolution of 2011 as the country experienced a severe drop in tourism income and foreign direct investment, as stated by the International Monetary Fund (IMF) [2]. Egypt faces challenges in providing sufficient resources of energy sources, particularly petroleum and natural gas, which

According to the Middle East Economic Survey (MEES) [4], the electrical power generation capacity in Egypt as of May 2015 was approximately 31.5 GW, where about 88.4% of the electricity is fed in Egypt by fossil fuels and the remainder are fed by petroleum and renewable energy (mostly hydroelectric) as shown in Figure 1 [5].

The Egyptian cabinet on 09/17/2014 has approved the issue of the feed-in tariffs of electricity produced from renewable energy sources (wind and solar). The aim of the issue of the new tariff is the production of about 4,300 megawatts and includes this goal the production of 300 megawatts of small solar plants installed below 500 kW, 2000 MW of large size photovoltaic (PV) installations, with range between 500 kW and 50 MW and the residual 2 GW is from wind farms with capacities ranging from 20 MW to 50 MW [3]. The law had committed the

radiation scaled of 5.93 (kWh/m²/day), annual average wind velocity for the location is 4.92 m/Sec and average biomass supplying 105 tons per day. The proposed grid connected solar/wind/Biomass hybrid system is an efficient method to cut emissions and the investment of the energy system is not increased.

Index Terms: Renewable energy; FIT (feed-in tariff); Viability study; HOMER; PV/wind/Biomass hybrid system.

I. INTRODUCTION

totaled to a reliable 95% of the total Egypt's energy needs. A set of studies demonstrate that even though Egypt have a reserve of primary energy resources; it will face a deficit to cover the increasing demand for energy. Not surprisingly the government is attempting to diversify its energy sources to fill the fuel shortage, which could make Egypt always been a net importer of fuel, which affects the reserves foreign exchange in the central bank [3].

electricity distribution companies to buy electricity made from renewable energy plants through the Power Purchase Agreement (PPA), and prices declared for a period of twenty-five years for PV projects and twenty years for wind projects.

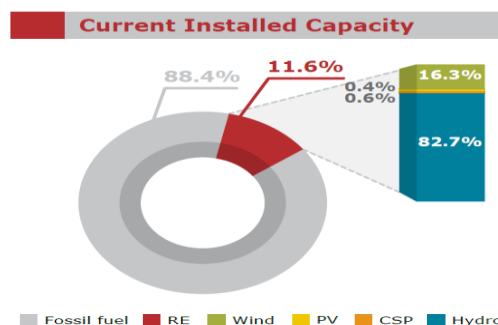


Figure 1 Egypt power production plan, including renewable resources

A contribution of the Egyptian government in resolving the persisting electricity crisis in the country, it has been encouraging citizens to invest in solar energy, by lending them for the establishment of stations and units of solar energy to generate electricity on the rooftop of their houses according to the new Feed-in Tariff consumers of electricity turns into producer and the difference between the electricity sale price for the government, and the high price for the purchase of which, is the most important incentive for residents of homes to install solar panels.

The rustic population strongly depends on agriculture, and then widely uses of conventional biomass are achieved. Renewable energy sources such as PV, wind and water are amply obtainable. Another major comparative benefit of rustic areas is open spaces and distances that can be readily used to install renewable energy systems (RES). Hence, if possible to merge all these renewable resources, in an effective method to achieve all their needs, then the issue of rustic growth can be effectively met. There are many existing methods used to determine the number of the constituent units of the hybrid systems, based on different RES whether it was solar, wind or biomass. [6].

Compare the operational features and cost values of three systems; off-grid solar-wind-diesel, solar-wind-hydrogen and on-grid solar-wind-hydrogen energy system [7]. It seemed from the comparability between the various systems that on-grid solar-wind-hydrogen energy system had the lowest total net present cost and cost of energy, and followed by solar-wind-diesel and off-grid solar-wind-hydrogen system. While inspect the techno-economic sides of exchanging diesel generators and batteries of the system by hydrogen system as well as doing the sizing optimization and simulation results of both schemes [8]. The analyses results showed that the changed of diesel generator with hydrogen technologies are technically practical, but still not economically applicable until mitigates in the cost of hydrogen technologies are made in the future. A pre-feasibility study has been directed to provide electricity to an off-grid village in Nepal [9]. The answers of the analyses indicated that solar panels are the most promising technology for that hamlet. Three types of arrangement were tested for the electrification of the entire village: Decentralized SHS, centralized solar generation and Centralized diesel fueled Gen-set.

Nowadays, computer based system simulation packages are used with growing common such as PSIM, MATLAB, and HOMER. Hybrid Optimization Model for Electric Renewable (HOMER) software package is widely applied in obtaining possible implementation and configurations.

Analysis and modeling executed suggesting an optimal solution of a hybrid system of RES by using the HOMER package software for remote areas are introduced before [10 – 17]. The Hybrid power systems (HPS) mentioned in these papers include a collection of various energy sources like wind/battery, SOLAR/battery, wind/ SOLAR /battery, wind/SOLAR /diesel/battery. While suggested a hybrid power system consisting of SOLAR arrays, biogas engine and wind turbine to feed the power needs of a village in Kenya [18]. The results show that the HPS integrated with the biggest engine as

backup can be a good solution than using a diesel engine as a standby.

In present work, the Monshaet Taher village in the Beni-Suef governorate has been selected for the development of the integrated renewable energy system in Egypt. Beni-Suef is a significant agricultural trade position on the west bank of the Nile River with area of 7,169 km.

II. STUDY AREA AND ENERGY RESOURCES

Actually, the grid is available in the village, but the supply of power is not reliable for supplying electric power continuously. Taking the advantages of plentiful renewable energy resources that available, it will make the electricity more reliable. The total area of the village is about 1463 acres, cultivated area represents, according to a statement of the Ministry of Agriculture 1447 acres, representing 98.6% of the area of the village. The village is characterized by the cultivation of crops, tomatoes, onions, strawberries and wheat.

In Monshaet Taher village, there are about 450 households with 7000 of local people [19]. Farming is the dominant source of income for rural households in this region and the remaining primary income source is shop owner, petty trader and casual labor. The main objective of this paper is to design an optimal economic power RES that supplies the desired electric load needs of Monshaet Taher village. The models have been designed to supply an optimal system arrangement based on daily data for energy availability and load demands.

A. Load profiles of Monshaet Taher

The village consists of 450 houses and others public affairs. The average load of the village is approximately 3102 kWh/day (or 129 kW) with 236 kW peak and has a load factor of 0.548 (which equates to the average load divided by the peak load, the load factor is $129 \text{ kW} / 236 \text{ kW} = 0.548$).

B. Hourly load demand curves

Load demand curves have been performed by utilizing a logical assumption suggesting that load demand varies in time, depends on the inhabitant presence in a room; that is why load demand curves are irregular and quite choppy over the time. Figure 2 exhibits the typical profile of the electrical load which is used in the case study, while Figure 3 shows the Monthly average load of the study area.

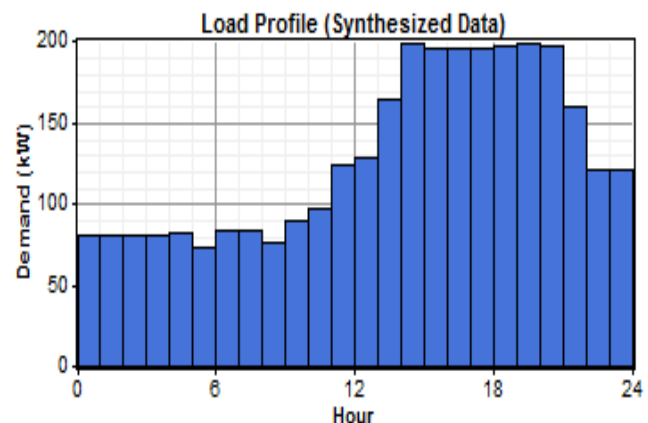


Figure 2: Daily load of the study area

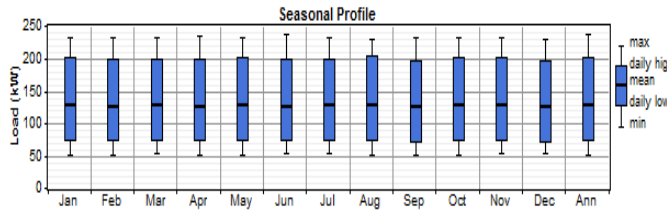


Figure 3: Monthly averages load of the study area

C. Renewable Energy Resources

1) Solar radiation:

The solar resource used for the Monshaet Taher village at a location of 29°2' N latitude and 31°6' E longitude was taken from the NASA Surface Meteorology and Solar Energy website [20]. The annual average solar radiation was scaled to be 5.93 kWh/m²/Day. The scaled data of global radiation and Clearness index of the study area are cleared in Figure 4.

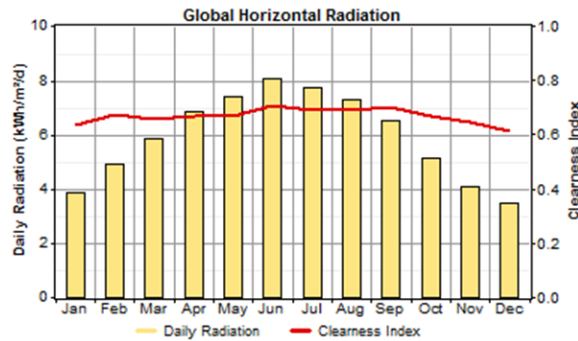


Figure 4: Solar energy profile in the study area.

2) Wind resource

The ten-year average monthly wind speed information was taken from the NASA website [20] based on the study area location. The annual average wind speed for the location is 4.92 m/sec. The scaled data of wind resource of the study area is shown in Figure 5.

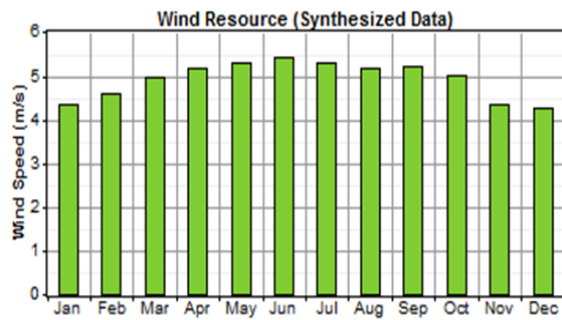


Figure 5: Wind energy profile at the selected village

3) Waste / Biomass

Every year during autumn there is the anniversary of the "black cloud", when farmers in the whole Nile-delta burn their agricultural waste, because they don't know what to do with it otherwise. This leads to a shrouded Cairo in a dark mixture of the usual traffic and industry exhausts and smog from e.g. burned rice straw. These unused biomass offers energy resources as well as the recycling and use of waste.

In 2010, Egypt generated 21 million tons of municipal solid waste (MSW) and the annual MSW generation growth is about 2%. Moreover, 6.2 million tons of industrial waste, 0.2 million tons of serious waste, 28,500 tons of medical waste, and about 25 million tons of agricultural waste were generated in the same year [21]. MSW contains 55-60% of organic waste. Because of its relatively high moisture content and its low calorific content, a Waste to Energy plant is compared to actual low electricity prices hardly feasible [21].

A biomass gasifier is considered during the present study. Using the produced biogas, a generator is involved in the Hybrid Renewable Energy (HRE) system to generate electricity. The average data of obtainable biomass feeding is about 105 tons / day [22]. The scaled data of Biomass resource of the study area is shown in Figure 6.

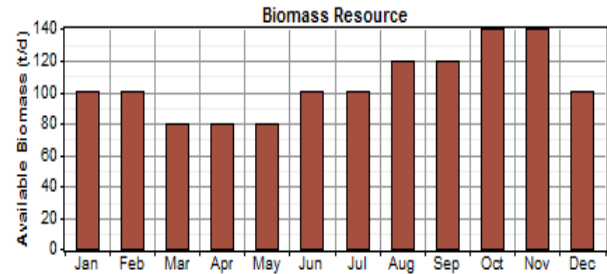


Figure 6: Biomass resources

D. Hybrid System Modeling

The optimal system combination will lead to the optimal system design with the lowest levelized cost of energy. HOMER package software application is applied to plan and correct financially and technically the choices for off-grid and on-grid power systems. HOMER can optimize the system configuration, and perform sensitivity analyses, and therefore, the designer can make the right decision when the supplying load is needed to optimize with minimization of energy cost.

The input data, including solar, wind and biomass resource data, electricity usage of the Monshaet Taher village and the components of solar/Wind/Biomass system are fed to the HOMER software tool built as a model plan shown in the following. Figure 7 depicts a general scheme of the proposed system in HOMER. The capital, replacement, operation and maintenance (O&M) costs of the various system elements have been introduced in Table 1.

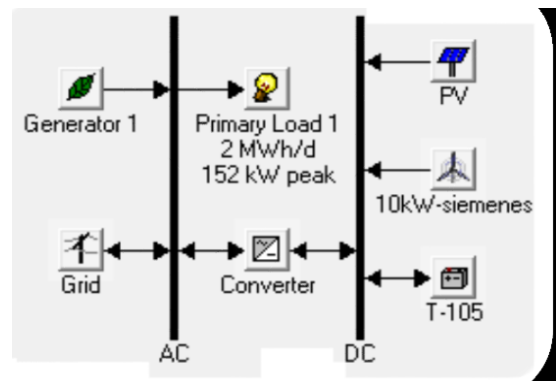


Figure 7: The optimum configuration model of the proposed system.

TABLE1: COST DETAILS OF THE VARIOUS EQUIPMENT USED IN THE PROPOSED SCHEME

		Capital (\$)	Replacement (\$)	O&M (\$/yrs.s)	Lifetime (yrs.s)
PV module	1 kW	1,500	1,350	10	25
Wind Turbine: Siemens 10 kW	1 piece	10,000	9,500	100	25
Battery: Trojan T-105	1 piece	200	180	5	Lifetime throughput 845 kWh
Biomass Generator	50 kW	20,000	18,000	1 \$/hr	15,000 h
Converter	3 kW	1,500	1,300	5	15
Grid	Power Price \$/kWh 0.038	Sellback Rate \$/kWh 0.122			

1) Emissions/Constraints:

The emissions and constraints in running the system are illustrated in Table 2.

TABLE2: EMISSIONS AND CONSTRAINTS OF THE SYSTEM

Carbon dioxide penalty	\$ 0/t
Peak annual capacity shortage	0, 3, 5%
Peak renewable fraction	0, 50, 75%
Operating reserve as percentage of hourly load	0.1
Operating reserve as percentage of maximum load	0
Operating reserve as percentage of solar power output	0.25
Operating reserve as percentage of wind power output	0.5

2) Simulation results:

Emulation was run in the HOMER software for several arrangements, in order to get the most effective system configuration that would give the lowest net present cost. Figure 8 presents all the various configurations and the results that can be acquired using the different RES.

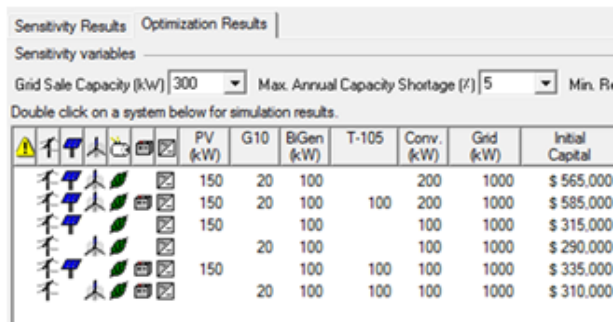


Figure 8 : The simulation results for PV – Wind – Biomass energy system

Grounding on the outcomes which is designated in Figure 8, HOMER has been able to enhance the energy efficiency of the system using different situations, by showing the results from the most cost efficient system to the minimum cost effective configuration. The summary of the optimal minimum

cost hybrid system for the proposed case study is illustrated in Tables 3, 4 and 5.

TABLE 3: SYSTEM ARCHITECTURE OF THE OPTIMIZED SYSTEM

PV Array	150 kW
Wind turbine	20 x 10kW- Siemens
Generator 1	100 kW
Grid	1,000 kW
Inverter	200 kW
Rectifier	200 kW

TABLE 4: COST SUMMARY OF THE OPTIMIZED SYSTEM

Total net present cost	\$ 5,772
Levelized cost of energy	\$ 0.000/kWh
Operating cost	\$ -43,747/year

TABLE 5: ELECTRICAL DETAILS OF THE OPTIMIZED SYSTEM

Component	Production (kWh/year)	Fraction
PV array	320,097	21%
Wind turbines	311,621	20%
Generator 1	774,000	51%
Grid purchases	122,370	8%
Total	1,528,088	100%

3) Details of Optimized Results:

The details of the optimized system costs are shown in Table 6.

TABLE 6: NET PRESENT COSTS OF THE OPTIMIZED SYSTEM

Component	Capital (1000\$)	Rep. (1000\$)	O&M (\$)	Salvage (\$)	Total (\$)
PV	225	0	19,175	0	244,175
10kW-siemenes	200	0	25,567	0	225,567
Generator 1	40	223	197,886	-839	460,514
Grid	0	0	-1,058,178	0	-1,058,178
Converter	100	36.16	4,261	-6,731	133,693
System	565	259.62	-811,288	-7,570	5,771

Figure 9 shows the monthly allocation of the electricity generated in kW by the proposed PV, Wind and Biomass generator.

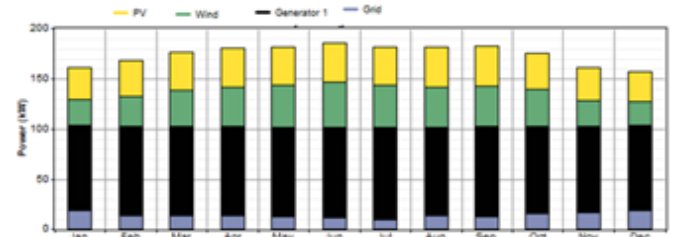


Figure 9: Monthly average electricity production from the optimized system.

It can be observed from Figure 9 and Table 7 that the Biomass generator is the dominant producer of electricity, for the chosen system. The Biomass generator operates for 7740 hours (capacity factor 88.4%), produces 774,000 kWh per year, with total rated capacity of 100 kW. The PV panels produce 320,097 kWh/year, with total rated capacity of 150 kW, operating for 4,384 hours/year (capacity factor of 24.4 %). The levelized cost of solar electricity is 0.0597 \$/kWh. While the 10kW-siemens Wind Turbines produce 311,621 kWh/year, with total rated capacity of 200 kW, operating for 8,760 hours/year (capacity factor of 42.7 %). The levelized cost of wind electricity is 0.0566 \$/kWh.

The optimized system produced 1,448,832 kWh/ year 50% (730,001) of it fed to the AC primary load of the village while the other 50% (718,830) of total electricity generated sales back to the grid. This demonstrates that this system has the ability in facing also the demand development in the future.

TABLE 7: TECHNO-ECONOMIC COMPARISON BETWEEN THE BEAT THREE HYBRIDS SYSTEMS

Configuration	Unit	Best optimize system	2 nd optimize system	3 rd optimize system
Solar PV	kW	150	150	150
Wind Turbine (10kW-siemens)	no.	20	20	0
Biomass generator	kW	100	100	100
Batteries (Trojan T-105)	no.	0	100	0
Biomass generator (Hours of operation)	hr/year	7740	7740	7118
Converter	kW	200	200	100
Grid	kW	1000	1000	1000
Emissions (Carbon dioxide)	kg/year	-376,833	-376,833	-158,255
Renewable Fraction	%	0.92	0.92	0.82
Total Capital Cost	\$	565,000	585,000	315,000
Total NPC	\$	5772	45,730	119,193
Levelized cost of energy	\$/kWh	0.000	0.000	0.000
Operating cost	\$/year	-43,747	-42,185	-15,317

Table 8 displays the Greenhouse Gases (GHG) emissions of the optimized system. Notice from the table that the emissions of the pollutants relevant to the network have a negative value.

TABLE 8: GREENHOUSE GASES (GHG) EMISSIONS

Pollutant	Emissions (kg/ year)
Carbon dioxide	-376,833
Carbon monoxide	4.89
Unburned hydrocarbons	0.541
Particulate matter	0.368
Sulfur dioxide	-1,634
Nitrogen oxides	-756

It can be observed from table 9 that the proposed optimized system is better and more economical than the grid connected system. From an environmental point of view, in terms of Greenhouse Gases (GHG) Emissions, the proposed hybrid PV/wind/biomass system is preferred over the grid connected system.

TABLE 9: COMPARISON BETWEEN THE PROPOSED OPTIMIZED SYSTEM AND GRID CONNECTED SYSTEM:

Item	the proposed optimized system	grid connected system
Configuration	PV/Wind/Biomass generator	Grid only
Total Capital Cost	\$ 565,000	\$ 0
Total NPC	\$ 5772	\$ 354,611
Levelized cost of energy	0.000 (\$/kWh)	0.038 (\$/kWh)
Operating cost	-43,747 (\$ /year)	27,740(\$ /year)
Carbon dioxide Emissions	-376,833 (kg/ year)	461,361(kg/ year)

E. Conclusions

HRE Systems are going to play a key role as the global electricity networks slowly minimize their dependence on fossil fuels. This is especially valid in the case of developing countries including Egypt.

The case study represents a typical grid-connected HRE system to feed a small village in Egypt. HOMER software package was used to compute the cost of energies produced by the hybrid system consists of solar photovoltaic, biomass and wind sources. The most optimal hybrid system is chosen based on environmental impacts and economic sides.

The results show the least-cost combination of HRE system that can face the demand in a reliable method at a cost of \$0.000/kWh and Total NPC of \$ 5772. Given the availability of solar and wind power at this location, most of the electricity in the optimal solution comes from the biomass generator and it provides a cheap source of power to the locality. From the simulation results (emissions and costs), it is clear that the use of hybrid SOLAR PV/wind / biomass system fulfills significantly lower NPC and mitigating on CO₂, as compared to a grid connected system. It is found that RES are practical method to solve the rural demand needs of Egypt.

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